

Research Question:

How does increasing password length affect the time required for a brute-force attack under different processing speeds?

Introduction

Nowadays, the world is in a digitally rising stage, where passwords serve as the first line of defense protecting personal and organizational data. Banking accounts, social media, billions of systems rely on passwords as the main authentication method. One of the simplest yet most persistent hacking techniques is the brute-force attack, in which every possible combination of characters is systematically tested until the correct password is found. As Bruce Schneier notes, *“Passwords that are too weak of course invite brute-force attacks.”* This statement emphasizes the mathematical nature of passwords and their security. The strength of the password depends not only on the secrecy of it but also on the enormous number of possible combinations an attacker must test.

A brute force attack is a hacking method that uses trial and error to crack passwords, login credentials, and encryption keys. It is a simple yet reliable tactic for gaining unauthorized access to individual accounts and organizations' systems and networks. The hacker tries multiple usernames and passwords, often using a computer to test a wide range of combinations, until they find the correct login information. A simple brute force attack occurs when a hacker attempts to guess a user's login credentials manually without using any software. A dictionary attack is a basic form of brute force hacking in which the attacker selects a target, then tests possible passwords against that individual's username.

In this investigation the character set size is $N = 62$, as throughout the investigation we are focusing on 26 lowercase, 26 uppercase letters and 10 digits. This investigation uses exactly this configuration as most usual passwords consist of this fragments. Using only this 62 characters will ensure that the analysis is clear and consistent.

Each character multiplies the number of possibilities by the size of the character set, and because of it the relationship between password length and cracking time is exponential. Understanding this exponential is critical for cybersecurity, as it quantifies how difficult a brute-force attack becomes as passwords grow longer. Cybersecurity masters are trying to make today's passwords more and more secure, even when we are registering for a new website or application we have the choice to use automatically generated passwords that are stronger. On the other hand where the security systems develop more protection, the speed of password-guessing is increasing with it. Relying on these websites (tomshardware.com, hivesystems.com, securitybrief.co.uk, hashcat.net, mojoauth.com) this is how the table for brute forcing looks alike for newest GPU and newest CPU, also the table shows brute-forcing with multiple GPUs of newest model.

Hardware	Attempts per Second (Approximate)
Single Nvidia RTX 5090	~740 million attempts/sec
12 Nvidia RTX 5090s (together)	~8.9 billion attempts/sec (approx.)
Intel Core i9-14900K	~2.9 million attempts/sec

(**Table 1.1** - The speed of GPU, 12 same GPUs together, CPU)

These findings raise an important question: *how secure are today's passwords in the face of accelerating computational speed?*

To model brute force attack and show the time that is required to hack it I will use a mathematical equation which looks like this.

$$T = \frac{N^L}{R}$$

Where

T is the total time to try all the possible combinations of the passwords

N is the number of possible characters in specific position

L is the password overall length

R is the rate of guesses in one second

This model is based on worldwide brute force password cracking standard theory, where all possible combinations must be tested sequentially. (Fortinet; Kaspersky)

By varying L and R, this exploration shows how increasing password length influences the time required for computers to hack using brute force in different computational technologies. The capital letters are used to represent constant configurations and the variables used in the model and investigation, which follows standards of mathematical notation.

Therefore, the research question guiding this study is: "How does increasing password length affect the time required for a brute-force attack under different processing speeds?"

I will aim to explore this topic through exponential and logarithmic modeling

The aim of this exploration is to use exponential and logarithmic modeling to compute this relationship, compare the effects of CPU, GPU, and quantum speeds , and evaluate the implications for future password security.

The numbers throughout the investigation will be rounded to three significant figures.

Rationale / Aim

I chose this topic because of my interest in cybersecurity and a part of mathematics that works with digital protection systems. We are interacting with passwords on a daily basis, but yet there are just few people who understand how they work, their mathematical principles, and how secure they actually are. I am planning to study computer science and that is why I wanted to explore the concept of passwords from mathematical perspectives and not just technical. Brute force attacks can be modelled using exponential functions due to the fact that they rely entirely on the number of possible combinations.

The aim of this investigation is to mathematically model how password length will affect the time that brute force attack needs to crack it under different processing speeds.

Data and Assumptions

In order to be able to model the time that is required for brute force attack, the necessary thing is to establish all the variables, constants, and assumptions clearly at the beginning of this investigation. This makes sure that all the results from this inspection will stay consistent, reproducible, and logically connected to real world cybersecurity situations. The model that has been used during this investigation is theoretical and all values are justified based on information that has been collected from 10 of October 2025 to 10 of November 2025. All the information will be provided correctly relying on the information that everybody could have access to during the period pointed above. This Internal Assessment will be justified based on modern hardware capabilities or reasonable projections of future technology.

Variables and Constant

Symbol	Meaning	Value(s) Used	Justification
N	Number of possible characters in each position (character set size)	62 (26 lowercase letters, 26 uppercase letters, 10 digits)	Represents a standard alphanumeric password format commonly used across websites and devices
L	Password length (number of characters)	4 to 14	Covers realistic user password lengths; upper bound shows exponential growth clearly
R_{CPU}	Attempt rate of a high-end consumer CPU	2.9×10^6 attempts/sec	Based on modern CPU benchmark values for password hashing or guessing
R_{GPU}	Attempt rate of a single modern GPU	7.4×10^8 attempts/sec	Represents a one of the highest level graphics cards; GPUs are widely used for password cracking
$R_{12\text{GPU}}$	Attempt rate of 12 GPUs working in parallel	8.9×10^9 attempts/sec	Shows the effect of distributed cracking rigs and parallel computing
R_{Quantum}	Theoretical future quantum computer speed	10^{16} attempts/sec	Represents a hypothetical high-performance quantum system for reflection purposes
T	Total time required to test all possible combinations	Calculated using the model	Dependent variable explored through the investigation

(Figure 1.2 - The variables used throughout the modelling process)

Assumptions of the model

To make sure that the investigation is controlled, the following assumptions are made:

1. Passwords are fully random and uniformly distributed

The characters of the password can be any N characters with equal probability. In the real world users rarely are using randomly created passwords, but this assumption is correct for pure brute force modelling.

2. Each hardware device at a constant rate R

CPU, GPU, and quantum speeds will stay constant throughout the attack. This will make it easier to model and will separate the effect of password length.

3. There are no system protections or lock out mechanisms

Real world applications and websites usually have mechanisms that limit the amount of login attempts or delay responses, but brute force attacks on password hashes face no such barriers.

4. No hashing, salting, or real world overhead is considered

The model focuses only on raw guessing speed, and not on cryptographically hashed delays, which will ensure the relationship between L and T is purely mathematical.

5. Parallel GPU cracking is modelled linearly

For 12 GPUs, the rate R is scaled directly by 12. This approximation is a standard for independent parallel tasks.

6. The half of the worst case time is the expected case time

Usually the attacker finds the correct password after checking half of the all possible combinations. Therefore, both worst case

$$T_{worst} = \frac{N^L}{R}$$

and expected case

$$T_{avg} = \frac{N^L}{2R}$$

are included.

Model Foundation

This investigation relies on an idea that a password that has length L and password length size N has:

$$Total\ combinations = N^L$$

When the device is able to try R passwords in one second, it means that the worst case time that is needed to test all the combinations is:

$$T = \frac{N^L}{R}$$

This formula stands as a backbone for my investigation. During this inspection this model will be more expanded and explained, will be linearised using logarithms, and will be tested across variety of values of R to illustrate the influence of password length in practical work of brute force attacks in both current and future computational environments.

Mathematical Modelling

In order to understand how password length is affecting the time that is required for a brute force attack, it is essential to first establish the structure that will describe the possible password number and the rate that different devices can test. The total number of combinations grows as the password length increases that is why brute force attacks follow exponential behavior. In this section will be developed a foundation that is clear and structured and will introduce exponential relationships, worst case and average case modelling and logarithmic linearization.

This section will start by modelling the number of possible passwords that might be created using the set of size N values independently, and the unique combinations numbers can be modelled as:

$$NL$$

The password length strongly influences security and this exponential expression is the reason for that. As an example, $N = 62$ and $L = 4$, the total number that represents the possible passwords will be $62^4 = 14,776,336$. If we will increase the length to 8 which will be $L = 8$, it will become $62^8 = 218,340,105,584,896$. This growth shows how impractical it will be to use brute force for longer passwords.

Converting this number into time, we will use R which is the number of attempts that device can make per second. In the worst case scenario we will get the password as the last option while testing, which means that to calculate the time needed to test all the combinations will be modelled like this:

$$T = \frac{NL}{R}$$

This is the equation that forms the central mathematical model of this investigation. It is combining the password length (L), the size of character set (N), and the speed of the hardware (R) to the time required for cracking (T). This investigation compares CPUs, and GPUs, and

also the theoretical quantum computer, which means that this formula directly helps to answer the research question.

As we found the formula for the worst case scenario, it is necessary to also show the more realistic scenario which is the average or expected cracking time. And since the correct password is equally likely to be anywhere between first and the last combinations, the time expected for half of the worst case scenario will be:

$$T_{avg} = \frac{NL}{2R}$$

Throughout this investigation two values will be calculated. For graphing the worst case scenario value will be used, as it emphasizes the exponential behaviour more clearly.

When L increases, the values are quickly becoming more large, which is making it harder to display or interpret directly. As an example 62^{12} and 62^{14} are making values that exceed 10^{21} . Logarithms will be used for the worst case model in order to make the exponential growth easier to analyse. Logarithms with base 10 will be used for both sides which will give:

$$\log_{10}(T) = \log_{10}(NL) - \log_{10}(R)$$

By using the logarithmic power rule, in simplified version it will look like this:

$$\log_{10}(T) = L \times \log_{10}(N) - \log_{10}(R)$$

This is a linear equation in L . $\log_{10}(N)$ is the slope of the line, which plays a role of constant for all devices as the set of characters stays the same at $N = 62$. The y-intercept depends on the value of R . When the device is more powerful and faster it will result in a lower intercept as the overall time will be reduced. The transformation above is essential as it allows the behaviour of T to be visualised as a straight line graph.

The sample calculations that are demonstrated below show how the formulas above are applied using real hardware speeds.

Sample Calculation 1(CPU, $L = 8$)

Firstly we are computing the values of combinations:

$$62^8 = 218,340,105,584,896$$

The next step is implementing the worst case scenario model:

$$T = \frac{62^8}{(2.9 \times 10^6)}$$

This will give approximately 7.51×10^7 seconds, which is 2.38 years.

This test showed that even very powerful CPUs nowadays will take a few years to crack even 8 character passwords.

Sample Calculation 2(GPU, $L = 10$)

Computing the total combinations needed:

$$62^{10} = 839,299,365,868,340,224$$

Worst case:

$$T = \frac{62^{10}}{(7.4 \times 10^8)}$$

This becomes approximately 1.13×10^9 seconds, which is approximately 36 years.

Sample Calculation 3 (Quantum Computer, $L = 10$)

Worst case:

$$T = \frac{62^{10}}{(10^{16})}$$

This will become approximately 83.9 seconds which is approximately 1.4 minutes.

This is a direct demonstration of how future quantum hardware could critically lower the times that are needed now to do it, and also the passwords that are used nowadays are considered as secure, but with quantum computing we can see that they are not.

These examples demonstrate how the mathematical model collaborates exponential growth with nowadays hardware performance in order to find the cracking time when brute force is used. Why is length the most important factor that affects the password? The answers are exponential relationships.

Data Tables

To be able to explore how password length affects the time required for brute force attacks while trying them on different hardware it is important to make a set with calculated values using the mathematical model developed earlier. Below are the values that were computed using the formulas.

$$T = \frac{NL}{R}$$
$$T_{avg} = \frac{NL}{2R}$$

The set of characters size stays the same at $N = 62$.

The speeds that are used for hardware are:

$$R_{CPU} = 2.9 \times 10^6 \text{ attempts per second}$$

$$R_{GPU} = 7.4 \times 10^8 \text{ attempts per second}$$

$$R_{12GPU} = 8.9 \times 10^9 \text{ attempts per second}$$

$$R_Q = 10^{16} \text{ attempts per second}$$

I have chosen to complete this investigation using password lengths from 4 till 14 as in real world examples usually have the length password that long, and also it illustrates the exponential growth.

All the tables contain:

- Total combinations
- Worst case time (T)
- Expected time (T_{avg})
- $\log_{10}(T)$ for the linearised graph.

All the values are rounded approximately like in the real world.

Total Number of Combinations(N^L)

L	62^L (Total combinations)
4	1.48×10^7
5	9.16×10^8
6	5.68×10^{10}
7	3.52×10^{12}
8	2.18×10^{14}
9	1.35×10^{16}
10	8.39×10^{17}
11	5.20×10^{19}
12	3.23×10^{21}
13	1.99×10^{23}
14	1.23×10^{25}

(Figure 1.3 - The demonstration of correlation of time and length)

These values already show exponential patterns. As an example, the jump from $L = 10$ to $L = 14$ is adding to the combinations by 10^7 times.

CPU Brute Force Times($R = 2.9 \times 10^6$)

$R_{CPU} = 2.9 \times 10^6$ attempts per second

The worst case time will be:

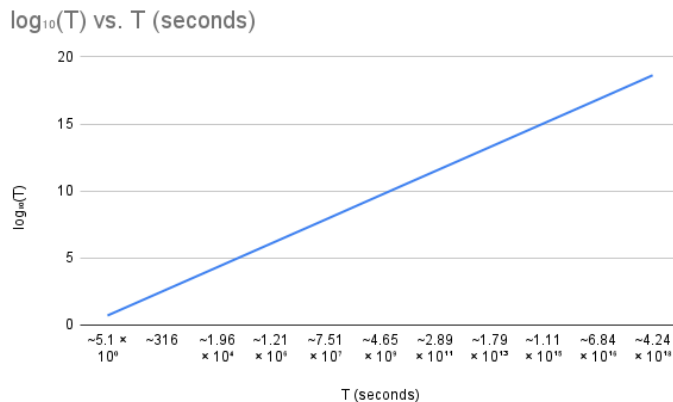
$$T_{worst} = \frac{NL}{R_{CPU}}$$

The expected time will be:

$$T_{avg} = \frac{NL}{2R_{CPU}}$$

L	T (seconds)	T _{avg} (seconds)	log ₁₀ (T)
4	~5.1 × 10 ⁰	~2.5	0.71
5	~316	~158	2.5
6	~1.96 × 10 ⁴	~9.8 × 10 ³	4.29
7	~1.21 × 10 ⁶	~6.0 × 10 ⁵	6.08
8	~7.51 × 10 ⁷	~3.75 × 10 ⁷	7.87
9	~4.65 × 10 ⁹	~2.32 × 10 ⁹	9.67
10	~2.89 × 10 ¹¹	~1.44 × 10 ¹¹	11.46
11	~1.79 × 10 ¹³	~8.97 × 10 ¹²	13.25
12	~1.11 × 10 ¹⁵	~5.54 × 10 ¹⁴	15.04
13	~6.84 × 10 ¹⁶	~3.42 × 10 ¹⁶	16.83
14	~4.24 × 10 ¹⁸	~2.12 × 10 ¹⁸	18.63

(Figure 1.3 - Password length versus cracking time)



(Graph1 - Graph of CPU cracking time vs length of the password)

Graph2 shows a growth of the time when the character length increases with each additional character multiplied by the total of possible passwords by 62, which is increasing the cracking time according to the $T = \frac{N^L}{R}$ model.

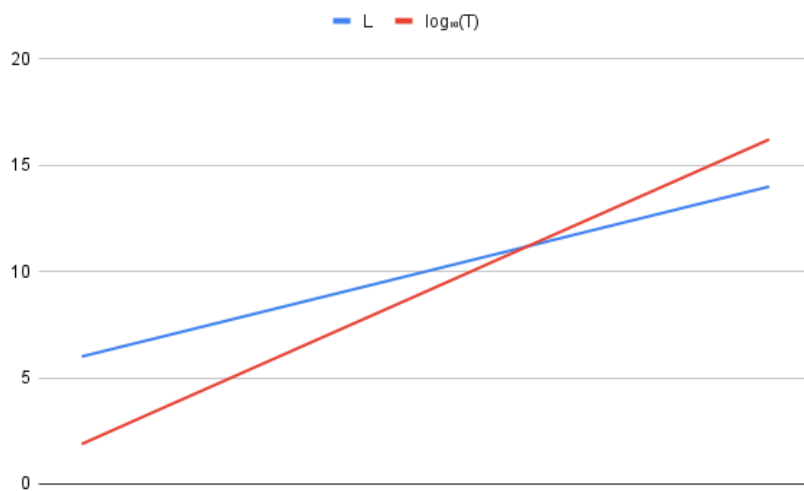
When the password length is short for example 4 or 5 characters according to the graph and table above the cracking time will take seconds or minutes, but when the characters length reaches 8 the cracking time rapidly increases up to several years and even higher for longer passwords. This shows that even with fixed processing speed, when the length of password is increased the secureness is improved.

GPU Brute Force Times($R = 7.4 \times 10^8$)

$R_{GPU} = 7.4 \times 10^8$ attempts per second.

L	T (seconds)	T_{avg} (seconds)	$\log_{10}(T)$
6	$\sim 7.6 \times 10^1$	$\sim 3.8 \times 10^1$	1.88
7	$\sim 4.76 \times 10^3$	$\sim 2.38 \times 10^3$	3.68
8	$\sim 2.95 \times 10^5$	$\sim 1.48 \times 10^5$	5.47
9	$\sim 1.83 \times 10^7$	$\sim 9.16 \times 10^6$	7.26
10	$\sim 1.13 \times 10^9$	$\sim 5.65 \times 10^8$	9.05
11	$\sim 7.0 \times 10^{10}$	$\sim 3.5 \times 10^{10}$	10.85
12	$\sim 4.37 \times 10^{12}$	$\sim 2.18 \times 10^{12}$	12.64
13	$\sim 2.69 \times 10^{14}$	$\sim 1.34 \times 10^{14}$	14.43
14	$\sim 1.68 \times 10^{16}$	$\sim 8.4 \times 10^{15}$	16.22

(Figure 1.4 - Password length versus cracking time for GPU)



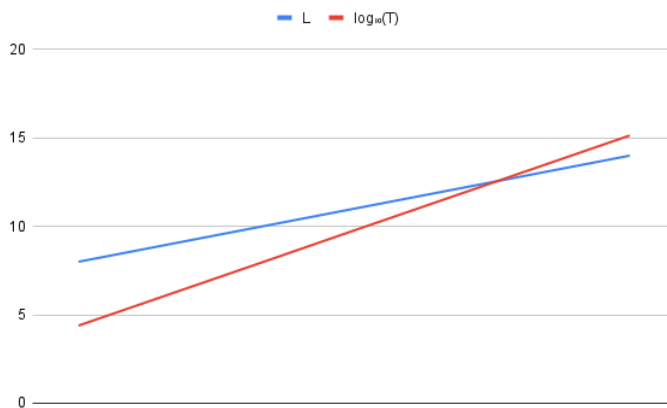
(Graph2 - GPU cracking time)

Multi-GPU Cluster(12 GPUs, $R = 8.9 \times 10^9$)

$R_{12GPU} = 8.9 \times 10^9$ attempts per second.

L	T (seconds)	T _{avg} (seconds)	log ₁₀ (T)
8	$\sim 2.45 \times 10^4$	$\sim 1.22 \times 10^4$	4.39
9	$\sim 1.52 \times 10^6$	$\sim 7.6 \times 10^5$	6.18
10	$\sim 9.43 \times 10^7$	$\sim 4.72 \times 10^7$	7.97
11	$\sim 5.85 \times 10^9$	$\sim 2.9 \times 10^9$	9.77
12	$\sim 3.63 \times 10^{11}$	$\sim 1.82 \times 10^{11}$	11.56
13	$\sim 2.25 \times 10^{13}$	$\sim 1.12 \times 10^{13}$	13.35
14	$\sim 1.39 \times 10^{15}$	$\sim 6.9 \times 10^{14}$	15.14

(Figure 1.5 - Password length versus cracking time with multiple GPUs)



(Graph3 - Multi GPU length vs time)

The GPU and multi GPU graphs have the same exponential patterns as the CPU but they have significantly lower cracking time due to their faster processing speed. One GPU is able to try passwords hundreds of times faster than CPU, but the overall the shape of the graphs remain the same.

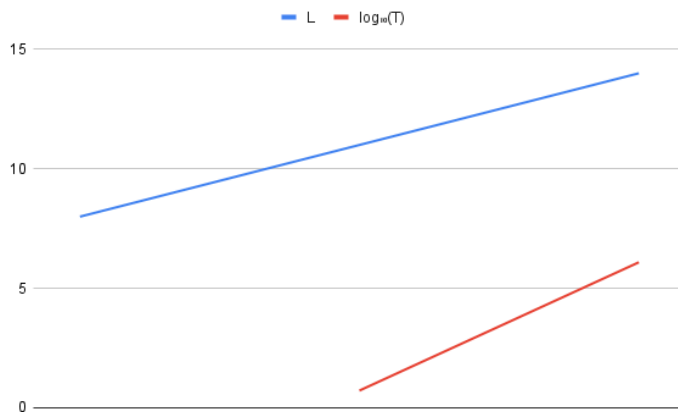
The multi GPU trial is increasing the attempt rate R but at the same time the exponential growth caused by increasing L stays dominant. This shows that even with higher speed of processing it only changes the graph vertically, and not changing the nature of exponents. This indicates that even when having higher speed of processing it will decrease the time of cracking passwords for short passwords but when the passwords are longer they are ineffective due to exponential dependence on password length.

Quantum Computer ($R = 10^{16}$)

$R_Q = 10^{16}$ attempts per second.

L	T (seconds)	T_{avg} (seconds)	$\log_{10}(T)$
8	$\sim 2.18 \times 10^{-2}$	$\sim 1.09 \times 10^{-2}$	-1.66
9	$\sim 1.35 \times 10^{-1}$	$\sim 6.8 \times 10^{-2}$	-0.87
10	$\sim 8.39 \times 10^{-1}$	$\sim 4.2 \times 10^{-1}$	-0.08
11	$\sim 5.20 \times 10^0$	~ 2.6	0.72
12	$\sim 3.23 \times 10^2$	$\sim 1.62 \times 10^2$	2.51
13	$\sim 1.99 \times 10^4$	$\sim 9.95 \times 10^3$	4.3
14	$\sim 1.23 \times 10^6$	$\sim 6.15 \times 10^5$	6.09

(Figure 1.6 - Password length versus cracking time for Quantum Computer)



(Graph4 - Quantum computer length vs time)

The graph shows that the length of the password has to be more than 10 characters for cracking time to be a positive number.

The graph of quantum computers shows how computers with very high processing speed would extremely shorten the time for brute-force attacks for shorter passwords. Passwords below 10 characters are negative numbers, while passwords that are $L < 10$ could be cracked in time less than a second depending on this theoretical model.

However with this configuration and cracking speed of 10^{16} the length of the password is this crucial part of the process and time. This shows that even with very fast cracking time the length of passwords remains the dominant fragment in security even for future computational

scenarios. This result states that longer passwords are the most important parts for defense strategy against brute force attacks.

Logarithmic Analysis

$$\log_{10}(T) = L \log_{10}(N) - \log_{10}(R)$$

The logarithms with base 10 are used to transform the worst case time exponential relationship into a linear one. The result will look like`

$$\log_{10}(T) = L \log_{10}(62) - \log_{10}(R)$$

This shows that the logarithm of cracking time is increasing linearly when the password length is also increasing. As the character set size is not changing in any device the the slope of the line remains constant with $\log_{10}(62)$ which means that all the devices produce straight with parallel lines on a $\log_{10}(T)$ versus L graph.

Y-intercept relies on the processing speed R , which shows that faster devices reduce the intercept but they do not affect the slope. This confirms that having a longer password mathematically is more effective to increase the time for brute force attack and therefore improving password security than making processing power faster.

Comparison of Processing Speeds

If we compare all devices on a single logarithmic scale we will see that as faster is processing speed as much it will reduce the cracking time but at the same if we increase the password length it will affect more to the time. For each character added to the password it will increase cracking time by the same multiplicative factor for any device used. This directly answers the research question by showing that password length is the factor that affects more on the brute force resistance rather than improving the processing speed. Even when quantum computer would be used for brute force attacks short passwords will be at risk while longer passwords are secure even under extreme numbers of trials.

Conclusion

This investigation shows how increasing password length affects the time required to crack it using brute force cyber attack type under different processing speeds using the model below.

$$T = \frac{N^L}{R}$$

The results suggest that the most important fragment in password guessing is the length of the password, while the increasement of processing speed reduces the linear reductions.

Throughout the investigation we used different scenarios including CPU, GPU, mulit GPU and a theoretical quantum computer and at the same time using the same fundamental behaviour. While adding just one character to the password and increasing the length of it the cracking time is multiplied by a constant factor determined by the character set size, at the same time increasing the processing speed changes the cracking time but only vertically and the exponential growth remains the same. Furthermore, the logarithmic linearisation showed that devices produce parallel straight lines and slopes, which shows that the length of the password is the most important thing in brute force attack defence.

During the investigation were used theoretically created quantum computing that significantly reduced the speed of cracking time, but at the same time for longer passwords we can see that it does not really short amount of time. This suggests that even against future computers and future configurations for password cracking using brute force attack, increasing the password length is more reliable than relying on limitations in computational power.

The model used during these investigations is using fully random passwords, constant guessing speeds and absence of real world protections such as hashes, salting, and rate limiting, this means that the theory has limitations. However this limitation are the key path to isolate the password length and cracking time and analyze them clearly.

Overall, this investigation shows that exponential growth plays the most important role in password security. While improving the processing time and reducing the cracking time we can see that even one character added to the password makes it hundreds of times harder to crack the password through brute force attack, which means that longer passwords are the most effective ways to defend against brute force attacks.

Bibliography

“What Is a Brute Force Attack? Definition, Types and How It Works | Fortinet.” *Fortinet*,

www.fortinet.com/resources/cyberglossary/brute-force-attack.

“Brute Force Attacks.” *Kaspersky.com*,

www.kaspersky.com/resource-center/definitions/brute-force-attack.

---. “Brute-force Attack.” *Wikipedia*, 24 Nov. 2025,

en.wikipedia.org/wiki/Brute-force_attack.

“NVIDIA GeForce RTX 5090 Graphics Cards.” *NVIDIA*,

www.nvidia.com/en-eu/geforce/graphics-cards/50-series/rtx-5090.

“NVIDIA GeForce RTX 5090 Specs.” *TechPowerUp*,

www.techpowerup.com/gpu-specs/geforce-rtx-5090.c4216.

Shifts, MojoAuth Pulse-Identity Stories. Industry. “How Fast Can RTX 5090 GPUs Crack

Passwords? A 2025 Benchmark.” *MojoAuth Pulse - Identity Stories. Industry*

Shifts, 17 Sept. 2025,

mojoauth.com/news/how-fast-can-rtx-5090-gpus-crack-passwords-a-2025-benchmark.

Klotz, Aaron. “Nvidia RTX 5090 Can Crack an 8-digit Passcode in Just 3 Hours

&Mdash; Password Cracking Benchmarks Show Tremendous&Hellip;” *Tom’s*

Hardware, 9 May 2025,

www.tomshardware.com/pc-components/gpus/nvidia-rtx-5090-can-crack-an-8-digit-passcode-in-just-3-hours.

“Intel Core i9 14900K.” *intel.com*,

www.intel.com/content/www/us/en/products/sku/236773/intel-core-i9-processor-14900k-36m-cache-up-to-6-00-ghz/specifications.html.

Mujtaba, Hassan. “Intel 14th Gen Core i9-14900K, Core i7-14700K, Core i5-14600K

CPUs Official: Same Prices as 13th Gen With More Performance.” *Wccftech*, 16

Oct. 2023,

wccftech.com/intel-14th-gen-core-i9-14900k-core-i7-14700k-core-i5-14600k-cpus-official-same-prices-more-performance.

Carter, Ethan. "GPU Password Cracking Benchmarks 2025: RTX Vs CPUs." *Online*

Hash Crack, 1 Dec. 2025,

www.onlinehashcrack.com/guides/password-recovery/gpu-password-cracking-benchmarks-2025-rtx-vs-cpus.php.